

Gesture AI Studios - Evaluation Report

Project Name: Gesture AI Studios

Description: Gesture AI Studios is an AI-powered Smart Mirror application that enables administrators to create customizable gesture-based interactive experiences using computer vision and hand tracking. Users interact naturally through hand gestures while the system displays customized messages, colors, and animations in real time.

1. Test Environment

Item	Value
Camera	Laptop Webcam
Framework	Streamlit
Backend	OpenCV + MediaPipe
Programming Language	Python
Operating System	Windows 11
Testing Date	26 June 2026

2. Gesture Test Results

Gesture	Expected Result	Actual Result	Status
Closed Fist	Display custom message	Works correctly	Pass
One Finger	Display custom message	Works correctly	Pass
Peace Sign	Display custom message	Works correctly	Pass
Three Fingers	Display custom message	Works correctly	Pass
Open Hand	Menu/Action	Works correctly	Pass
Rock Sign	Display custom message	Occasional misclassification	Needs Improvement
Index & Pinky	Display custom message	Works correctly	Pass
Thumbs Up	Recognize thumbs up	Occasional misclassification	Needs Improvement
Thumbs Down	Recognize thumbs down	Occasional misclassification	Needs Improvement

3. Performance Evaluation

- Average startup time: 2–3 seconds.
- Average gesture response time: Less than 0.2 seconds.
- Estimated frame rate: Approximately 20–30 FPS depending on lighting and hardware.
- System stability: Stable during continuous testing.

4. Screenshots

Insert screenshots of:

- Administrator Panel
- Smart Mirror
- Hand Detection
- Customized Gesture
- GitHub Repository

5. Accuracy Observations

The application performs reliably for clearly distinguishable gestures such as Closed Fist, Open Hand, Peace Sign, and Three Fingers. Similar gestures, particularly Thumbs Up, Thumbs Down, and Four Fingers, may occasionally be misclassified due to the current rule-based gesture recognition approach. Future versions will incorporate a custom machine learning gesture classifier to improve accuracy.

6. Challenges

- Similar hand poses can be difficult to distinguish.
- Lighting conditions affect recognition quality.
- Camera quality influences overall detection performance.

7. Future Improvements

- Train a custom machine learning gesture classification model.
- Support image, video, and audio actions.
- Add gesture sequences and workflow automation.
- Deploy to Streamlit Community Cloud or Hugging Face Spaces.
- Support multiple users and saved projects.

8. Pictures:

